

Monomial $f(q) = a \cdot q^6$ Report

Polynomial

$$f(q) = aq^6$$

Naive

$$g_{\text{naive}}(X) = X^6 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 720a \frac{\Delta^6}{\epsilon} + 360a \frac{\Delta^4}{\epsilon} q^2 + 30a \frac{\Delta^2}{\epsilon} q^4 + aq^6$$

$$\begin{aligned} \text{Var}(g_{\text{naive}}(X)) &= 478483200a^2 \frac{\Delta^{12}}{\epsilon} + 238982400a^2 \frac{\Delta^{10}}{\epsilon} q^2 + 19785600a^2 \frac{\Delta^8}{\epsilon} q^4 \\ &\quad + 642240a^2 \frac{\Delta^6}{\epsilon} q^6 + 10260a^2 \frac{\Delta^4}{\epsilon} q^8 + 72a^2 \frac{\Delta^2}{\epsilon} q^{10} \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{naive}}) &= 479001600a^2 \frac{\Delta^{12}}{\epsilon} + 239500800a^2 \frac{\Delta^{10}}{\epsilon} q^2 + 19958400a^2 \frac{\Delta^8}{\epsilon} q^4 \\ &\quad + 663840a^2 \frac{\Delta^6}{\epsilon} q^6 + 11160a^2 \frac{\Delta^4}{\epsilon} q^8 + 72a^2 \frac{\Delta^2}{\epsilon} q^{10} \end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 720a \frac{\Delta^6}{\epsilon} + 360a \frac{\Delta^4}{\epsilon} q^2 + 30a \frac{\Delta^2}{\epsilon} q^4$$

Unbiased

$$g_{\text{unb}}(X) = X^6 a - 30X^4 a \frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^6$$

$$\begin{aligned} \text{Var}(g_{\text{unb}}(X)) &= 297561600a^2 \frac{\Delta^{12}}{\epsilon} + 148780800a^2 \frac{\Delta^{10}}{\epsilon} q^2 + 12398400a^2 \frac{\Delta^8}{\epsilon} q^4 \\ &\quad + 413280a^2 \frac{\Delta^6}{\epsilon} q^6 + 7380a^2 \frac{\Delta^4}{\epsilon} q^8 + 72a^2 \frac{\Delta^2}{\epsilon} q^{10} \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{unb}}) &= 297561600a^2 \frac{\Delta^{12}}{\epsilon} + 148780800a^2 \frac{\Delta^{10}}{\epsilon} q^2 + 12398400a^2 \frac{\Delta^8}{\epsilon} q^4 \\ &\quad + 413280a^2 \frac{\Delta^6}{\epsilon} q^6 + 7380a^2 \frac{\Delta^4}{\epsilon} q^8 + 72a^2 \frac{\Delta^2}{\epsilon} q^{10} \end{aligned}$$

Comparison

$$\text{mean gap} = -720a \frac{\Delta^6}{\epsilon} - 360a \frac{\Delta^4}{\epsilon} q^2 - 30a \frac{\Delta^2}{\epsilon} q^4$$

$$\begin{aligned} \text{variance gap} = & -180921600a^2 \frac{\Delta^{12}}{\epsilon} - 90201600a^2 \frac{\Delta^{10}}{\epsilon} q^2 \\ & - 7387200a^2 \frac{\Delta^8}{\epsilon} q^4 - 228960a^2 \frac{\Delta^6}{\epsilon} q^6 - 2880a^2 \frac{\Delta^4}{\epsilon} q^8 \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = & -181440000a^2 \frac{\Delta^{12}}{\epsilon} - 90720000a^2 \frac{\Delta^{10}}{\epsilon} q^2 \\ & - 7560000a^2 \frac{\Delta^8}{\epsilon} q^4 - 250560a^2 \frac{\Delta^6}{\epsilon} q^6 - 3780a^2 \frac{\Delta^4}{\epsilon} q^8 \end{aligned}$$

$$\begin{aligned} \text{Bias}(g_{\text{naive}})^2 = & 518400a^2 \frac{\Delta^{12}}{\epsilon} + 518400a^2 \frac{\Delta^{10}}{\epsilon} q^2 \\ & + 172800a^2 \frac{\Delta^8}{\epsilon} q^4 + 21600a^2 \frac{\Delta^6}{\epsilon} q^6 + 900a^2 \frac{\Delta^4}{\epsilon} q^8 \end{aligned}$$